

Linear Transformations

Read Chapt 3 in Notes!

Linear transformations are very special functions (maps) from a vector space V to a vector space W .

Definition Let V and W be vector spaces.

A function $T: V \rightarrow W$ is a linear transformation (linear map) if

1. For all $\mathbf{u}, \mathbf{v} \in V$,

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \text{ and}$$

"T preserves addition"

2. For all $\mathbf{v} \in V$ and scalars λ ,

$$T(\lambda \mathbf{v}) = \lambda T(\mathbf{v}).$$

"T preserves scalar multiplication"

addition in vector space V

addition in vector space W

scalar multiplication in V

scalar multiplication in W

A linear transformation T "respects"/preserves the linear space structure underlying V and W :

- add two vectors in V and then transform with T
= first transform each of the vectors with T and then add them in W
- take a vector in V , multiply with a number and then transform with T
= first transform the vector with T and then multiply with the number

It is common (tradition) to write linear transformations with capital letters.

Example Let $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$.

Then for any $m \times n$ matrix A , the function defined by

$$T_A(\mathbf{x}) = A\mathbf{x}$$

is a linear transformation.

Let's check this:

$$T_A(\mathbf{x} + \mathbf{y}) = A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = T_A(\mathbf{x}) + T_A(\mathbf{y}), \checkmark$$

$$T_A(\lambda \mathbf{x}) = A(\lambda \mathbf{x}) = \lambda A\mathbf{x} = \lambda T_A(\mathbf{x}). \checkmark$$

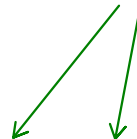
Each matrix A defines a linear transformation in this way!

We will see later that each linear transformation (on finite-dimensional vector spaces) can be represented by a matrix.

Concrete example 1

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

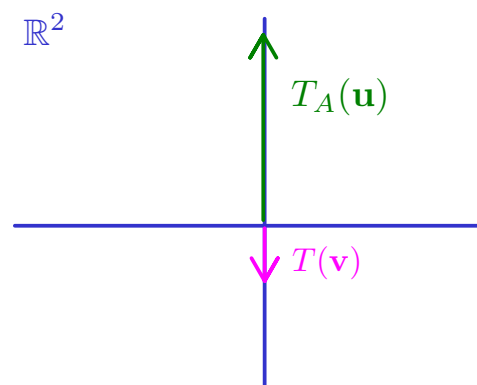
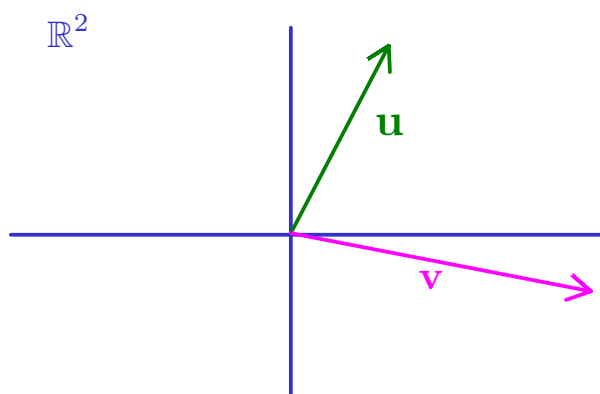
Implicit in this notation: Operations addition and scalar multiplication are the usual ones. (Simply Entry-wise addition, etc.)



Then T_A is a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$:

For any $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2$,

$$T_A \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ x_2 \end{pmatrix}.$$



T_A is a projection on the y-axis.

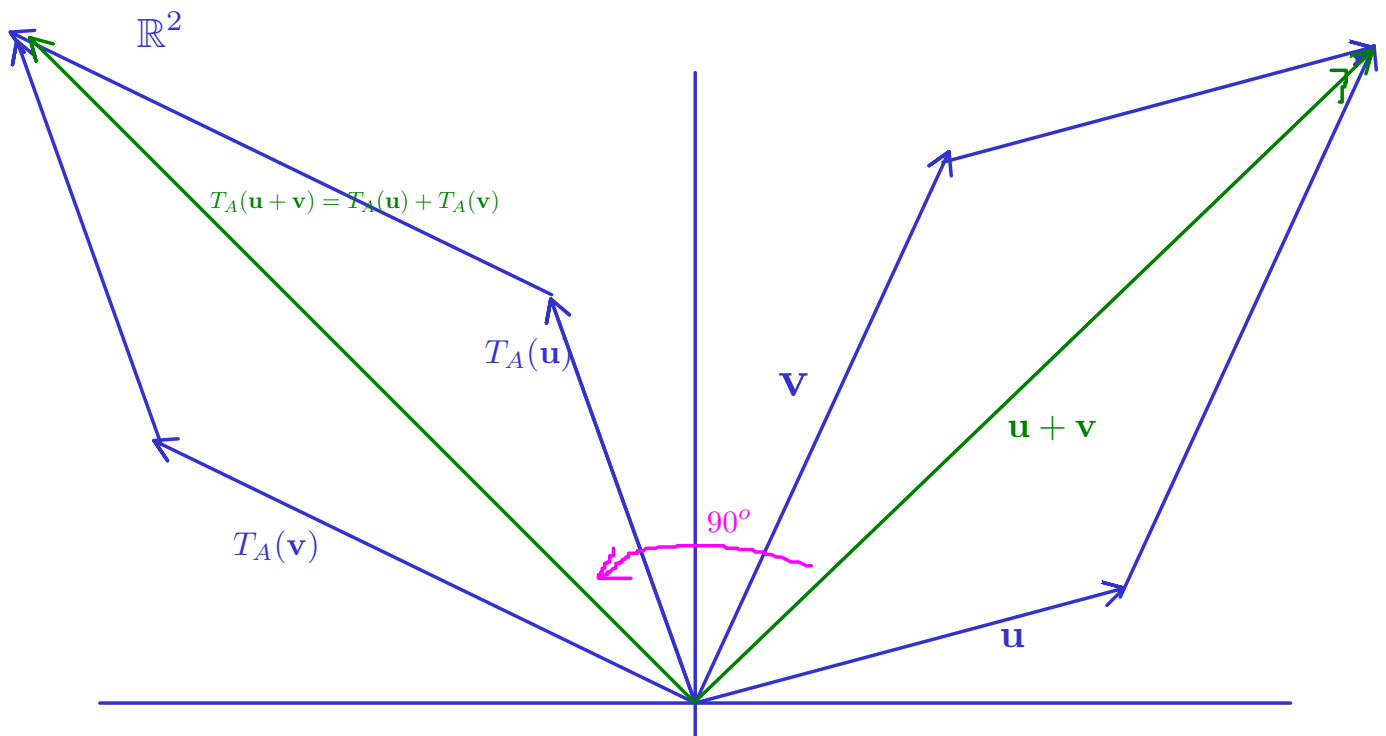
Concrete example 2

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Then T_A is a linear transformation that is a rotation through an angle θ .

Let's for instance take $\theta = 90^\circ$. Then

$$T_A \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \underset{A}{\begin{pmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{pmatrix}} \underset{x}{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}} = \underset{A}{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}} \underset{x}{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}} = \underset{T_A(x)}{\begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}}.$$



In this figure, we also see that the rotation preserves addition:

Rotating u and v by 90 degrees and then adding them gives the same result as first adding u and v and then rotating by 90 degrees.

You can verify the same for scalar multiplication.

Example Let $V = P_3$ and $W = P_2$.

Define the function $T: P_3 \rightarrow P_2$ by

$$T(p(x)) = p'(x).$$

T preserves addition:

$$T(p(x) + q(x)) = (p(x) + q(x))' = p'(x) + q'(x) = T(p(x)) + T(q(x))$$

T preserves scalar multiplication:

$$T(\lambda p(x)) = (\lambda p(x))' = \lambda p'(x) = \lambda T(p(x))$$

Hence T is a linear transformation.

Example $T: P_1 \rightarrow P_3$ defined by

$$T(ax + b) = x^2(ax + b)$$

is a linear transformation.

Check this!

Example $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$

is not a linear transformation.

For instance, we have

$$T\left(2\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right) = T\left(\begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

while

$$2T\left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right) = 2\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}.$$

not equal

Hence T does not preserve scalar multiplication. (One counter example is enough.)

Therefore, T is not a linear transformation.

But the following is a linear transformation!

We only changed 1 to 0 here.

Example $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} \right) = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$ (0 is special in this situation.)

is a linear transformation.

For instance, we have

$$T \left(2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right) = T \left(\begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \right) = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$$

while

$$2T \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right) = 2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}.$$

equal!

This is just an example and does not prove that T preserves addition (left as an exercise).

You may also check that T preserves scalar multiplication.

Some general properties of linear transformations

Theorem Let $T : V \rightarrow W$ be a linear transformation.

Then

1. $T(\mathbf{0}) = \mathbf{0}$.
zero vector in V zero vector in W
2. For any $\mathbf{v} \in V$,
 $T(-\mathbf{v}) = -T(\mathbf{v})$.

Proof 1. Let $\mathbf{v}$ be any vector in V .

Then

$$T(0\mathbf{v}) = T(\mathbf{0})$$

because in any vector space, $0\mathbf{v} = \mathbf{0}$.

On the other hand,

$$T(0\mathbf{v}) = 0T(\mathbf{v}) = \mathbf{0}.$$

because T preserves scalar multiplication.

Together, these equations imply that $T(\mathbf{0}) = \mathbf{0}$.

2. $T(-\mathbf{v}) = T((-1)\mathbf{v}) = (-1)T(\mathbf{v})$.

